

Revealed Preferences

JEB104 Microeconomics I
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Revealed Preference

Preferences not observable, we observe only choices.

What do choices reveal about consumer's preferences?

Are choices consistent with our assumptions on rationality of preferences?

Previously

- Choice of utility-maximizing individual derived from his preferences (represented by utility function) and budget constraint

Today

- Completely different approach to analysis of individual decision-making assuming that preferences are not observable
- We can however recover preferences based on observing choice

Observable choices

- We know prices, we know income, but we do not know map of indifference curves (preferences), it is so abstract that we cannot observe it.
- But we observe choices – chosen bundles of individuals under given prices and income.

We can use information about consumer's choices to discover his preferences.

In our previous analysis, we assumed that preferences are rational, which ensured that they are relatively reasonable (to have well-behaved preferences we even added a few more assumptions). So in this analysis, we also need individual choices to satisfy some consistency criteria.

Assumptions about Preferences

Preferences

- do not change while the choice data are gathered
- are strictly convex
- are monotonic

Strict convexity and monotonicity $\Rightarrow$ the preferred affordable bundle is **unique**

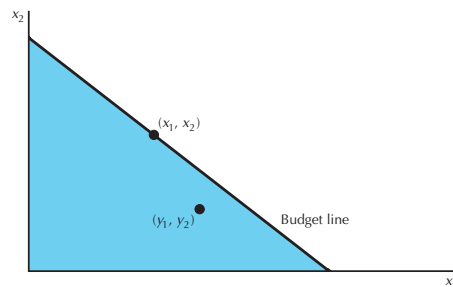
Before we proceed to the discussion on consistency criteria choices should satisfy, we have to make several assumptions on underlying preferences that generate individual choice.

Monotonicity implies that bundles containing more of some goods and not less of other goods are never worse and strictly convex preferences ensure that for given prices and income combination, optimal bundle is unique.

Now we will forget about utility functions for a while

- We will only look at what is observable: prices, income, chosen bundle

Direct Revealed Preference



Consumer chooses (x_1, x_2) and not (y_1, y_2) when both are available. $\Rightarrow$

$\Rightarrow$ The bundle (x_1, x_2) is **directly revealed preferred** to the bundle (y_1, y_2) .

Let $\mathbf{x}^0 = (x_1(p^0, m^0), x_2(p^0, m^0), \dots, x_n(p^0, m^0))$ and $\sum_{i=1}^n x_i^0 p_i^0 = m^0$, then for any $\mathbf{y}$, such that $\mathbf{y} \neq \mathbf{x}^0$ and $\sum_{i=1}^n y_i p_i^0 \leq m^0$,

$$\mathbf{x}^0 \succ^D \mathbf{y}.$$

Firstly, we start with the definition of direct revealed preference. So what does it mean if we say that one bundle is directly revealed preferred to another bundle?

Definition of “ $\mathbf{x}$ is directly revealed preferred to $\mathbf{y}$ ”

- $\mathbf{x}$ is chosen when $\mathbf{y}$ is available ($\mathbf{x}$ is chosen over $\mathbf{y}$)

$\mathbf{x}^0$ is the bundle which is chosen under prices $\mathbf{p}^0$ and income m^0

If $\mathbf{x}^0$ is the chosen bundle, we know that it brings our consumer the highest utility level possible given all the available bundles (bundles in the budget set), so:

- It must be preferred to all bundles under the budget line
- It must be preferred to all other bundles on the budget line because we assume that the optimal bundle is unique

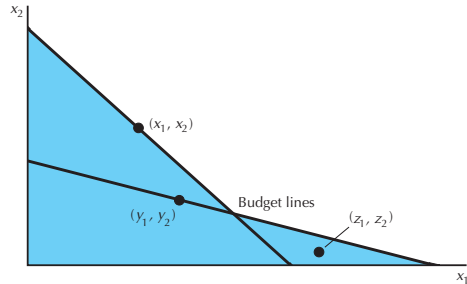
Formal definition: Let $\mathbf{x}^0$ be chosen bundle under prices $\mathbf{p}^0$ and income m^0 lying on the budget line. Then, for any other bundle $\mathbf{y} = (y_1, y_2, \dots, y_n)$ lying in the budget set, we must have that $\mathbf{x}^0$ is directly revealed to $\mathbf{y}$.

Note that: $\mathbf{x}^0 = (x_1(p^0, m^0), x_2(p^0, m^0), \dots, x_n(p^0, m^0)) = (x_1^0, x_2^0, \dots, x_n^0)$

Only the direct revealed preference would not help us much

- We need indirect as well

Indirect Revealed Preference



- Under (p^0, m^0) , consumer chooses (x_1, x_2) and not (y_1, y_2) when both are available.
- Under (p^1, m^1) , he chooses (y_1, y_2) and not (z_1, z_2) when both are available.

(x_1, x_2) directly revealed preferred to (y_1, y_2) , (y_1, y_2) directly revealed preferred to $(z_1, z_2) \Rightarrow (x_1, x_2)$ is **indirectly revealed preferred** to (z_1, z_2)

Holds for any length n : $x^0 \succ^D x^1, x^1 \succ^D x^2, x^2 \succ^D x^3, \dots, x^{n-1} \succ^D x^n \Rightarrow x^0 \succ^I x^n$.

Weak Axiom of Revealed Preference (WARP)

Weak axiom of revealed preference (WARP)

If bundle x is ever directly revealed preferred to bundle y , $x \neq y \Rightarrow y$ can never be directly revealed preferred to x :

$$x \succ^D y \Rightarrow y \not\succ^D x$$

- Choices violating the WARP are inconsistent with economic rationality.
- WARP is a **necessary** condition for economic rationality
- WARP is **not** a sufficient condition for economic rationality

There are bundles which cannot be compared directly (optimal bundle and a bundle lying above the respective budget set), but we can still say which one is better (and for this purpose we use indirect revealed preference)

Draw consecutively by yourself:

1. Budget line with x as the optimum and point z . We cannot judge which one brings consumer higher utility level.
2. Add the point y , y lies inside the original budget set, so $x \succ^D y$.
3. Now draw a budget line corresponding to the budget set, from which y is chosen as the optimum.
4. And compare y and z , z is available while y is chosen, hence $y \succ^D z$.
5. $x \succ^D y$ and $y \succ^D z$ imply that $x \succ^I z$, so x is indirectly revealed preferred to z .

Choice should be consistent.

- We want to study a rational consumer. We aim to find such an axiom imposed on choice that would ensure that preferences generating choice are rational.
- What criterion must the choice satisfy to be consistent? Weak axiom of revealed preferences.

WARP

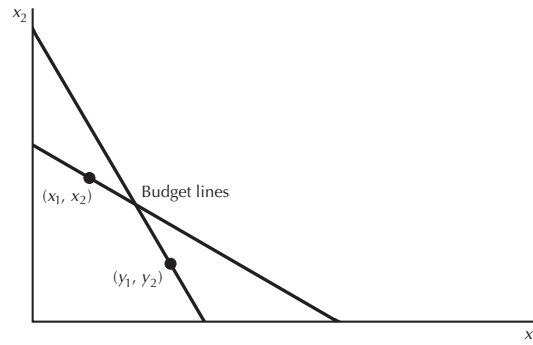
- Suppose there exists a prices-income combination under which bundle x is chosen, while y is available, so $x \succ^D y$ (y is inside the budget set or on the budget line, but x is optimum lying on the budget line). If there is a different price-income combination under which y is chosen (y is optimum lying on the budget line), then x cannot be affordable $y \not\succ^D x$ (x must be above the budget line).

WARP is **necessary**, but not **sufficient** condition for preferences to be rational

- Suppose that consumer has rational preferences and these preferences generate some choice behavior. Then, this choice behavior will be consistent with WARP.
- However, if we observe choice behavior that satisfies WARP, it does not mean that it was generated by rational preferences. (It may violate transitivity as you will see later in this lecture.)

Graphical examples showing choices violating and satisfying WARP follow.

Weak Axiom of Revealed Preference (WARP)



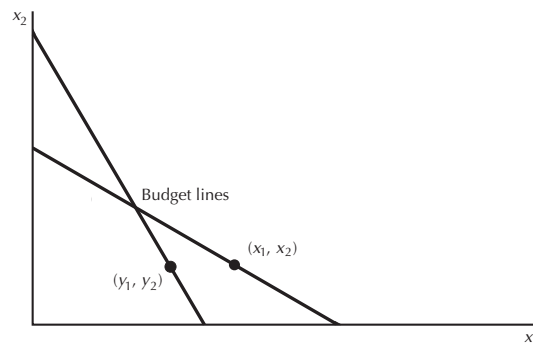
(x_1, x_2) chosen when (y_1, y_2) available $\Rightarrow (x_1, x_2) \succ^D (y_1, y_2)$

(y_1, y_2) chosen when (x_1, x_2) available $\Rightarrow (y_1, y_2) \succ^D (x_1, x_2)$

$\Downarrow$

Violation of WARP

Weak Axiom of Revealed Preference (WARP)



(x_1, x_2) chosen when (y_1, y_2) available $\Rightarrow (x_1, x_2) \succ^D (y_1, y_2)$

When (y_1, y_2) chosen, (x_1, x_2) not available

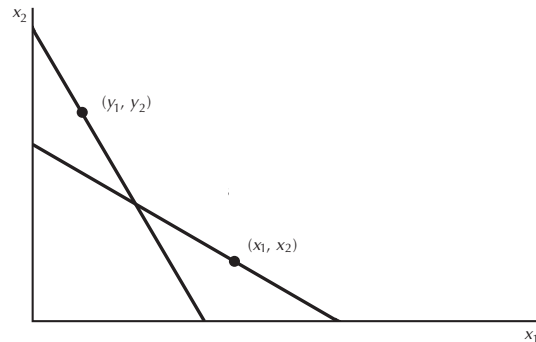
$\Downarrow$

WARP is satisfied

This choice behavior violates WARP, because in both cases one was chosen while the other one was available. So at one situation $A \succ B$, while in the other situation $B \succ A$.

This choice behavior is a perfect example of choice compliant with WARP. In this case, x was chosen over y , but when y was chosen, x was not available, so WARP is satisfied.

Weak Axiom of Revealed Preference (WARP)



When (x_1, x_2) chosen, (y_1, y_2) not available

When (y_1, y_2) chosen, (x_1, x_2) not available

We cannot compare bundles according to direct revealed preference.



Choice does not violate WARP (WARP is satisfied)

WARP: Exercise

A consumer makes the following choices:

1. $(p_1, p_2) = (\$2, \$2)$: $(x_1, x_2) = (10, 1)$
2. $(p_1, p_2) = (\$2, \$1)$: $(x_1, x_2) = (5, 5)$
3. $(p_1, p_2) = (\$1, \$2)$: $(x_1, x_2) = (5, 4)$

Is the WARP violated by these data?

Prices/Choices	1. (10,1)	2. (5,5)	3. (5,4)
1. (\$2, \$2)	\$22	\$20	\$18
2. (\$2, \$1)	\$21	\$15	\$14
3. (\$1, \$2)	\$12	\$15	\$13

If WARP is not violated, we can say that it is satisfied.

Choices – chosen bundle for given prices. And we ask whether this choice behavior satisfy Weak Axiom of Revealed Preference.

Firstly, we construct a table with expenditures on bundles for different prices

- Rows: different prices
- Columns: different chosen bundles
- diagonal – costs of chosen bundles – prices in situation i multiplied with bundles in situation i
- 1st row: $2 \cdot 10 + 2 \cdot 1 = 22$, $2 \cdot 5 + 2 \cdot 5 = 20$, $2 \cdot 5 + 2 \cdot 4 = 18$
- 2nd row: $2 \cdot 10 + 1 \cdot 1 = 21$, $2 \cdot 5 + 1 \cdot 5 = 15$, $2 \cdot 5 + 1 \cdot 4 = 14$
- 3rd row: $1 \cdot 10 + 2 \cdot 1 = 12$, $1 \cdot 5 + 2 \cdot 5 = 15$, $1 \cdot 5 + 2 \cdot 4 = 13$

Does income of our consumer stay the same in these three price situations?

No. We can compare numbers on diagonal – \$ 22, \$15 and \$13 are incomes our consumer has in each situation (income is fully spent on the goods considered).

WARP: Exercise

Prices/Choices	1. (10,1)	2. (5,5)	3. (5,4)
1. (\$2, \$2)	\$22	\$20	\$18
2. (\$2, \$1)	\$21	\$15	\$14
3. (\$1, \$2)	\$12	\$15	\$13

Black: costs of chosen bundles (incomes in different situations)

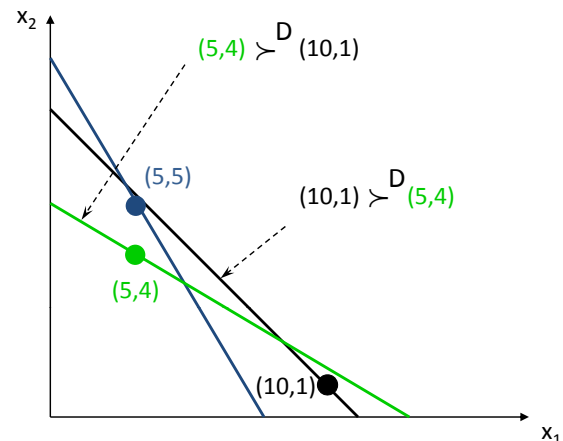
Green: bundles available under given p, m , but not chosen: Black $\succ^D$ Green

Red: bundles not available under given p, m : Cannot compare Black and Red

Prices/Choices	1. (10,1)	2. (5,5)	3. (5,4)
1. (\$2, \$2)	chosen	$(10,1) \succ^D (5,5)$	$(10,1) \succ^D (5,4)$
2. (\$2, \$1)		chosen	$(5,5) \succ^D (5,4)$
3. (\$1, \$2)	$(5,4) \succ^D (10,1)$		chosen

But $(10,1) \succ^D (5,4)$ and $(5,4) \succ^D (10,1) \Rightarrow$ The choice **violates** WARP.

WARP: Exercise



Now we can look at bundles which were affordable for each particular situation.

- If cost of bundle is lower than income (=cost of bundle that is chosen under given prices), then bundle is affordable. (GREEN COLOUR)
- If cost of bundle is higher than income (=cost of bundle that is chosen under given prices), then bundle is not affordable. (RED COLOUR)

For each row, we can then check direct revealed preference:

- 1st row: Bundle 1 $\succ$ Bundle 2 and Bundle 1 $\succ$ Bundle 3 (bundles 2 and 3 affordable, but bundle 1 chosen)
- 2nd row: Bundle 2 $\succ$ Bundle 3 (bundle 3 affordable, but bundle 2 chosen)
- 3rd row: Bundle 3 $\succ$ Bundle 1 (bundle 1 affordable, but bundle 3 chosen)

Now we will put direct revealed preferences altogether and we have that $1 \succ 3$ and at the same time, $3 \succ 1$, which violates WARP.

The previous exercise graphically

- We can clearly see that WARP is violated

- We could see it even better if the blue line (second prices combination) is not there

Slopes should be a bit different

WARP: Not Sufficient for Rational Preferences

- Choice $X \succ^D Y$, $Y \succ^D Z$ and $Z \succ^D X$ may satisfy WARP
- Non-transitive preferences = not rational
- WARP is **not sufficient condition** for having rational preferences.

If we have a non-transitive choice, it may satisfy WARP, but it will not be rational (remember that one of the rationality axioms is transitivity).

- We cannot draw it as it's impossible to do so with 2 goods (we need at least 3 goods to draw this situation).

Not sufficient $\Rightarrow$ We need a **stronger** condition $\Rightarrow$ SARP

Strong Axiom of Revealed Preference (SARP)

While WARP only focuses at direct revealed preferences, SARP has to hold for indirect revealed preferences as well.

WARP and SARP are equivalent if there are only two goods.

Strong axiom of revealed preference

If bundle x is ever revealed preferred to bundle y (directly or indirectly), $x \neq y$, y can never be directly or indirectly revealed preferred to x .

SARP stronger than WARP: SARP imposes transitivity on revealed choices

Even though the choice ($X \succ^D Y$, $Y \succ^D Z$ and $Z \succ^D X$) satisfies WARP, it does not satisfy SARP.

SARP: Exercise

A consumer makes the following choices:

- A: $(p_1, p_2, p_3) = (1, 3, 10)$, $(x_1, x_2, x_3) = (3, 1, 4)$
- B: $(p_1, p_2, p_3) = (4, 3, 6)$, $(x_1, x_2, x_3) = (2, 5, 3)$
- C: $(p_1, p_2, p_3) = (1, 1, 5)$, $(x_1, x_2, x_3) = (4, 4, 3)$

Is the SARP violated by these data?

Prices/Choices	A	B	C
A	\$46	\$47	\$46
B	\$39	\$41	\$46
C	\$24	\$22	\$23

SARP: Exercise

Prices/Choices	A	B	C
A	\$46	\$47	\$46
B	\$39	\$41	\$46
C	\$24	\$22	\$23

Same as previously: Black $\succ^D$ Green, cannot directly compare with Red

Prices/Choices	A	B	C
A	Chosen		$A \succ^D C$
B	$B \succ^D A$	Chosen	
C		$C \succ^D B$	Chosen

The choice **satisfies** WARP.

But $A \succ^D C$, $C \succ^D B \Rightarrow A \succ^I B \Rightarrow$ The choice **violates** SARP ($B \succ^D A$).

By the same logic: $B \succ^D A$, $A \succ^D C \Rightarrow B \succ^I C$ and $C \succ^D B$, $B \succ^D A \Rightarrow C \succ^I A$

It is easier to under the concept using an exercise.

Now we have 3 goods and 3 different price combinations. And we want to check whether this choice behavior satisfies SARP.

Let's calculate costs of bundles in the table again. You can check by yourselves that values are correct, now you know how to do it.

Again, we check which bundles were affordable (green) and which not (red), and we compare bundles using the direct revealed preferences.

This time the choice satisfies WARP

- There are no two bundles going "against each other", so there are no contradictions in mutual relationships.

But the choice behavior is not transitive: $A \succ^D C$, $C \succ^D B$ and $B \succ^D A$. Hence, the choice does not satisfy SARP.

- Bundle A is indirectly revealed preferred to B, while B is directly revealed preferred to A.

Strong Axiom of Revealed Preference

The choice satisfies SARP **if and only if** it is generated by rational preference relation.

That the observed choice data satisfy the SARP is a condition **necessary and sufficient** for there to be a well-behaved preference relation that “rationalizes” the choice data.

Therefore, if SARP is satisfied, we know that preferences are rational. And if preferences are rational, the generated choice behavior will be consistent with SARP (and also WARP).

Recovering Preferences

The theory of revealed preference can help us recover region where indifference curves may lie.

Suppose we observe the following consumer choice satisfying SARP:

$$A : (p_1, p_2) = (1, 1); (x_1, x_2) = (15, 15)$$

$$B : (p_1, p_2) = (2, 1); (x_1, x_2) = (10, 20)$$

$$C : (p_1, p_2) = (1, 2); (x_1, x_2) = (20, 10)$$

$$D : (p_1, p_2) = (2, 5); (x_1, x_2) = (30, 12)$$

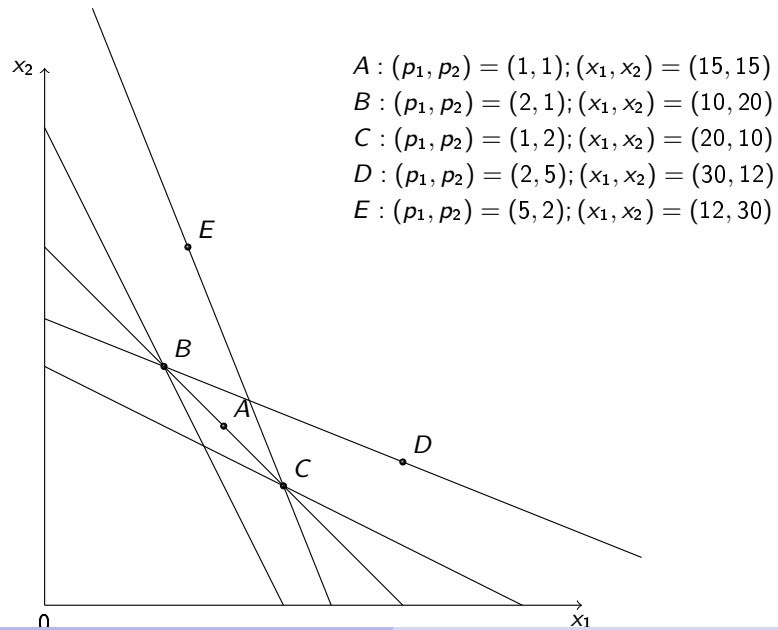
$$E : (p_1, p_2) = (5, 2); (x_1, x_2) = (12, 30)$$

How can we locate the indifference curve containing point A ?

We have 5 different budget sets and respective optimal bundles. What is the set of potential bundles that are indifferent to bundle A ?

We will explore it graphically.

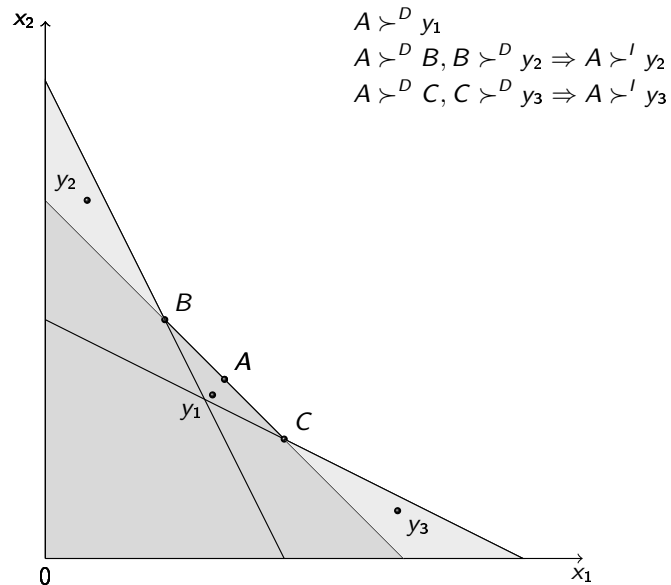
Graphical Representation



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Recovering Regions Below Indifference Curve



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We are interested in where the indifference curve which goes through point A lies.

We can analyze these regions using the revealed preference theory

Use an elimination method

- Find areas which have to lie below and above the indifference curve
- The indifference curve has to be between them

Let's start by the lower region

- All bundles that A is preferred to: $\forall y : A \succ y$
- Forget about points D and E now

y_1 and the darker-gray region is "dominated" by A

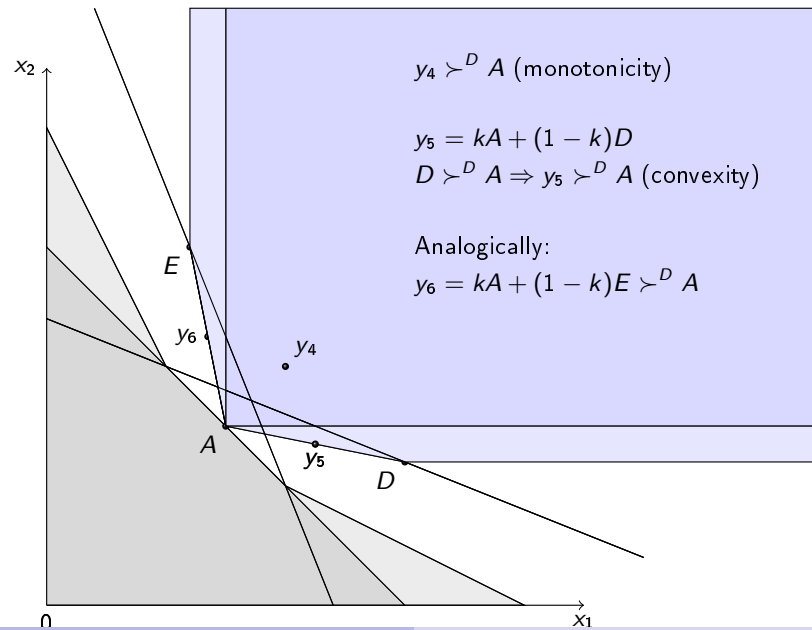
- This is straightforward as A is chosen from the darker-gray region which represents its respective budget set.

y_2 and y_3 and the light gray region are those where A is indirectly preferred

- This is where revealed preference helps, A is directly revealed preferred to B and C, because bundles B and C lie on the budget line of the budget set out of which A is chosen
- B is directly revealed preferred to y_2 (affordable, but B is chosen), C is directly revealed preferred to y_3 (affordable, but C is chosen), then A is indirectly revealed preferred to y_2 and y_3
- Note that we need SARP to hold

NEXT: Upper region

Recovering Regions Above Indifference Curve



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y_4 and the darker blue region is preferred to A

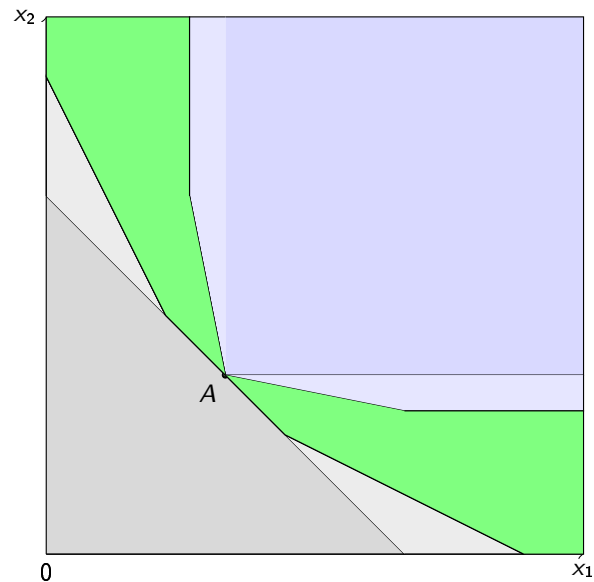
- This is a direct consequence of monotonicity (bundles containing more of both goods must be better)

y_5 and y_6 and the light blue regions are also preferred to A

- Because D and E are preferred to A (D and E are chosen while A is available), the convex combination of D/E and A must also be preferred to A (result of strict convexity of preferences)

NEXT: Simply combine the two

Indifference Curve Must Lie between the Two Regions

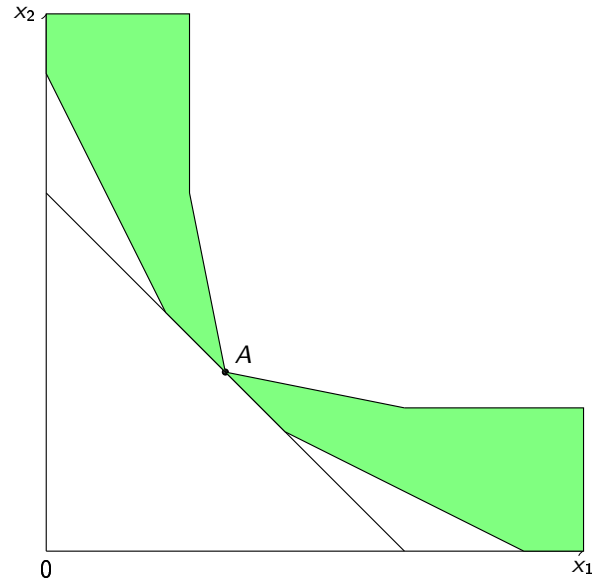


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Indifference curve must lie between the two regions

Revealed Preferences Helped Us to Narrow Down the IC Shape



This is how the potential region looks with the original budget set.

The more choices we observe, the narrower region for IC can be drawn. So this is the way how observation of choices helps us to recover shape of indifference curves.